STUDY PROJECT - PHASE 2

(Design & Proof of Concept)

BCSZC241T

Explainable Machine Learning Framework for Battery Health Assessment and Dynamic Backup-Time Prediction

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1. Introduction

1.1 Purpose of Phase 2

Phase 2 translates the approved Phase-1 problem definition into a concrete system design and demonstrates that the main technical approach is feasible. The phase defines the detailed functional and non-functional requirements, common data schema, software architecture, module interfaces, data flow, storage approach, model design, technology stack, and testing strategy. It also implements a limited Proof of Concept (PoC) that exercises the core data-to-prediction path before full integration and public-dataset validation in Phase 3.

The final project title remains unchanged from Phase 1: Explainable Machine Learning Framework for Battery Health Assessment and Dynamic Backup-Time Prediction. The Phase-2 design preserves the Phase-1 scope boundary: the core targets are remaining capacity/SOH and dynamic backup time; Remaining Useful Life, embedded BMS control, production deployment, and safety-critical decision authority remain outside the core Study Project scope.

1.2 Scope of Phase 2

- Detailed system and software architecture for the complete framework.
- Detailed functional and non-functional requirements derived from Phase 1.
- Common battery-data schema, source-adapter rules, target definitions, and leakage controls.
- Module-wise design for ingestion, validation, preprocessing, feature engineering, baseline calculation, ML prediction, evaluation, explainability, optional decision support, and dashboard reporting.
- Logical data-flow and artefact-storage design; a relational database is not required for the Phase-2 local prototype.
- Technology-stack selection and justification.
- A working controlled-benchmark PoC using 720 deterministic records, a battery-grouped development/holdout split, capacity and SOH prediction, formula verification, dynamic backup-time prediction, evaluation, decision-support checks, and exportable comparison graphs.
- Testing strategy, risk review, and explicit readiness criteria for Phase 3.

Phase 2 does not claim real-world battery accuracy from the Controlled Synthetic Golden Benchmark (CSGB). The benchmark verifies calculations, validation rules, leakage controls, model wiring, grouped evaluation, and reporting; it is not laboratory ground truth. Quantitative validation on held-out public experimental batteries remains a Phase-3 activity.

1.3 Traceability from Phase 1

Phase-1 Primary Objective	Phase-2 Design Mapping	Phase-2 Evidence
PO1 - Modular data framework	M1-M3; schema, validation, preprocessing and feature design	Architecture + common schema + PoC pipeline
PO2 - Capacity/SOH assessment	M5	Leakage-safe capacity model and SOH equation
PO3 - Dynamic backup-time prediction	M6	Separate load-aware Random Forest workflow
PO4 - Transparent formula baseline	M4 + M7	Formula baseline compared on common test records
PO5 - Evaluation and explainability	M7-M8	Grouped holdout metrics, actual-vs-predicted plots, residual/error graphs, M9 exact-match checks; SHAP/local explanation retained for Phase 3.
PO6 - Interactive software prototype	M10 with M1-M9 services	Streamlit architecture and interface specification; full dashboard integration carried to Phase 3

2. System Overview

2.1 Product Perspective

The proposed system is a standalone, local-first Python application designed as a modular analytical framework rather than as an embedded battery-management controller. Users interact with a Streamlit interface, while the application layer orchestrates source adapters, validation, feature engineering, formula calculations, machine-learning models, evaluation, explainability, and reporting. Public datasets or project-generated test files are treated as external inputs. Processed datasets, trained models, metrics, configuration values, and plots are stored as local artefacts so the project can be reproduced on a normal student laptop.

The design uses a layered architecture. The presentation layer contains the dashboard and user interaction. The application layer contains the functional modules and orchestration logic. The data/model layer contains standardized files, persisted preprocessing artefacts, trained models, metrics, and visual outputs. This separation permits later replacement of the local file system with a database or service without redesigning the model logic.

2.2 Major System Functions

- Import CSV or Excel battery data and map source-specific fields to a documented common schema.
- Validate required fields, numeric types, units, duplicates, missing values, and basic physical plausibility.
- Clean accepted records and generate leakage-safe electrical, thermal, cycle, energy, load, and degradation features.
- Derive a discharge-duration / backup-time target from time-series discharge data when the source dataset supports a valid derivation.
- Calculate a formula-based runtime baseline using predicted/available capacity, representative voltage, load power, and configurable efficiency.
- Train or load a capacity regressor and calculate SOH from predicted capacity and reference/rated capacity (Roman et al., 2021; Ren & Du, 2023; Khaleghi et al., 2024).
- Train or load a separate dynamic backup-time regressor (Tu et al., 2024; Wang & Chen, 2020).
- Compare ML and formula estimates on common evaluation records and calculate MAE, RMSE, MAPE, bias, and R^2 where valid.
- Produce global and local explanations using feature importance, permutation importance, and SHAP where supported (Lundberg & Lee, 2017; Rahmanian et al., 2024).
- Present validation feedback, predictions, comparison charts, explanation outputs, assumptions, and downloadable results through a dashboard.

2.3 User Classes and Characteristics

User / External Class	Characteristics and Interaction
Analyst / student / researcher	Loads datasets, reviews data quality, trains/evaluates models, interprets metrics and explanations. Expected to understand basic battery variables and model limitations.
Maintenance / operational viewer	Uses a selected record or valid operating-condition input to view SOH and backup-time estimates. The output is decision support only and does not replace field testing or safety procedures.
Project developer / maintainer	Maintains schema mappings, configuration, model artefacts, tests, dependencies, and code versions; can add dataset adapters or future algorithms.
External data source	NASA PCoE, CALCE, Oxford, or future datasets. These are not users but are treated as external systems supplying experimental records with source-specific formats.

3. Functional Requirements

The following requirements are carried forward from Phase 1 and refined into design-level responsibilities. The PoC-status column indicates only what is demonstrated by the Phase-2 synthetic implementation; it does not imply completion of public-dataset validation or the final Streamlit application.

ID	Functional Requirement	Module	Phase-2 Status
FR1	Import supported CSV or Excel battery data.	M1	Implemented for the CSGB Excel workbook; all 720 rows imported.
FR2	Map source-specific column names to the documented common schema where a valid mapping exists.	M1	Implemented through a benchmark-to-common-schema adapter; public-source adapters remain Phase 3.
FR3	Verify required fields, data types, units, and basic physical plausibility before training/prediction.	M2	Executed; 720/720 benchmark rows retained as valid.
FR4	Report missing, invalid, duplicate, and out-of-range records and apply documented preprocessing rules.	M2	Quality report generated; no silent row loss, null output, or duplicate record_id.
FR5	Generate leakage-safe electrical, thermal, cycle, load, energy, and degradation-related features.	M3	Demonstrated using leakage-safe cycle, electrical, thermal, load, resistance, C-rate, and voltage-sag features.
FR6	Derive discharge-duration/backup-time target from time-series discharge records where valid.	M1-M3	Algorithm defined; public-dataset implementation pending.
FR7	Calculate a configurable formula-based backup-time baseline.	M4	Implemented as an exact known-answer check and a fair operational comparator using M5-predicted capacity.
FR8	Train an initial Random Forest regression model for dynamic backup-time prediction.	M6	Implemented; trained on 540 development rows and evaluated on 180 grouped holdout rows.
FR9	Train a separate remaining-capacity regression workflow without capacity-derived leakage inputs.	M5	Implemented; capacity/SOH model trained without target columns and evaluated on unseen batteries.
FR10	Calculate predicted SOH from predicted capacity and rated/reference capacity.	M5	Implemented; predicted SOH calculated from predicted capacity and rated capacity.
FR11	Compare formula and ML backup-time estimates on the same evaluation records.	M7	Implemented on the same 180 held-out record IDs with zero battery overlap.
FR12	Calculate MAE, RMSE, MAPE, bias, and R^2 where mathematically valid.	M7	Implemented; MAE, RMSE, MAPE, bias, and R^2 exported for M4-M6 outputs.
FR13	Display actual-versus-predicted and residual/error visualizations.	M7/M10	Implemented; 18 PNG files are supplied (11 primary plus seven supplementary), representing 11 distinct analytical views.
FR14	Provide at least one global and one local explanation for ML predictions.	M8	Partially implemented: built-in and 10-repeat permutation importance provide two global M5/M6 explanation views; local explanation remains a Phase-3 integration item.
FR15	Generate SHAP explanations when model/package support is available.	M8	Optional SHAP helper and a separate pinned dependency file are supplied; SHAP was not installed for the submitted run and full local/global SHAP evidence remains Phase 3.
FR16	Dashboard shall accept valid new-condition or selected-record inputs and return health/runtime estimates.	M10	UI flow designed; full Streamlit integration carried to Phase 3.
FR17	Export cleaned data, predictions, metrics, and selected explanations.	M10	Implemented for Phase 2; 34 generated evidence files comprise 16 data/report/validation outputs and 18 PNGs. project_outputs_sample.csv is a documented 180 x 8 audit input and is not counted.
FR18	Save trained model and preprocessing artefacts for later reuse.	M5/M6/M10	Artefact design defined; final persistence integration carried to Phase 3.
FR19	Optional decision-support module maps SOH/runtime/load/data-quality indicators to configurable non-safety-critical guidance.	M9	Implemented and evaluated with exact-match accuracy, per-class precision/recall/F1/support, confusion matrices, and 100% reference-input rule wiring.

ID	Functional Requirement	Module	Phase-2 Status
FR20	Dashboard displays assumptions and limitations so outputs are not mistaken for certified safety or field-validation results.	M10	Implemented in report outputs; synthetic-data and non-safety-critical boundaries stated explicitly.

4. Non-Functional Requirements

4.1 Performance Requirements

- Validation and prediction should remain interactive for small and medium capstone datasets on a normal laptop.
- The initial Random Forest models must run without a GPU; training time is allowed to be longer than single-record inference.
- Persisted models should permit later predictions without mandatory retraining.
- For very large time-series inputs, the design shall permit chunking/batch processing rather than loading all records into the UI at once.

4.2 Security Requirements

- Input files are treated as data and must not be executed.
- Only supported file types and documented schemas are accepted by the prototype.
- The local Phase-2 design does not require user authentication; any future network/cloud deployment must add authentication, access control, upload-size limits, secure storage, and audit logging.
- Model outputs are labelled as decision support and must not be presented as certified battery-safety decisions.

4.3 Usability Requirements

- Use clear labels, physical units, and short explanatory text for all major inputs and outputs.
- Distinguish fatal validation errors from warnings that allow continued analysis.
- Allow normal prototype use without editing Python source code once the dashboard is integrated.
- Present predictions together with assumptions, metrics, visual comparisons, and explanation caveats.

4.4 Scalability and Maintainability

- Each functional module has a documented input/output contract so algorithms or data sources can be replaced independently.
- Source-specific dataset mapping is isolated in adapters rather than embedded throughout model code.
- Configuration values such as efficiency, column aliases, health thresholds, random seeds, and paths are centralized.
- Dependencies are declared in requirements.txt; code and documentation are version controlled in GitHub.
- The file-based artefact layer can later be replaced by relational or time-series storage without changing model mathematics.

4.5 Reliability and Reproducibility

- Invalid records must not silently enter training.
- Training and evaluation records remain separated; preprocessing learned from training data is applied consistently to test and inference data.
- Random seeds, dataset versions, schema mappings, metrics, and model artefacts are recorded for repeatable demonstrations.
- Target leakage controls are treated as a design constraint for both the capacity and backup-time workflows.

4.6 Explainability Requirements

- Provide at least one model-level explanation and one individual-prediction explanation in the final integrated implementation.
- Use multiple explanation views when practical (model importance, permutation importance, and SHAP) because correlated variables can alter attribution rankings; attribution is descriptive rather than causal (Lundberg & Lee, 2017).
- State clearly that feature attribution describes the model response and does not prove physical causation.

5. System Architecture and Design

5.1 System Architecture Diagram

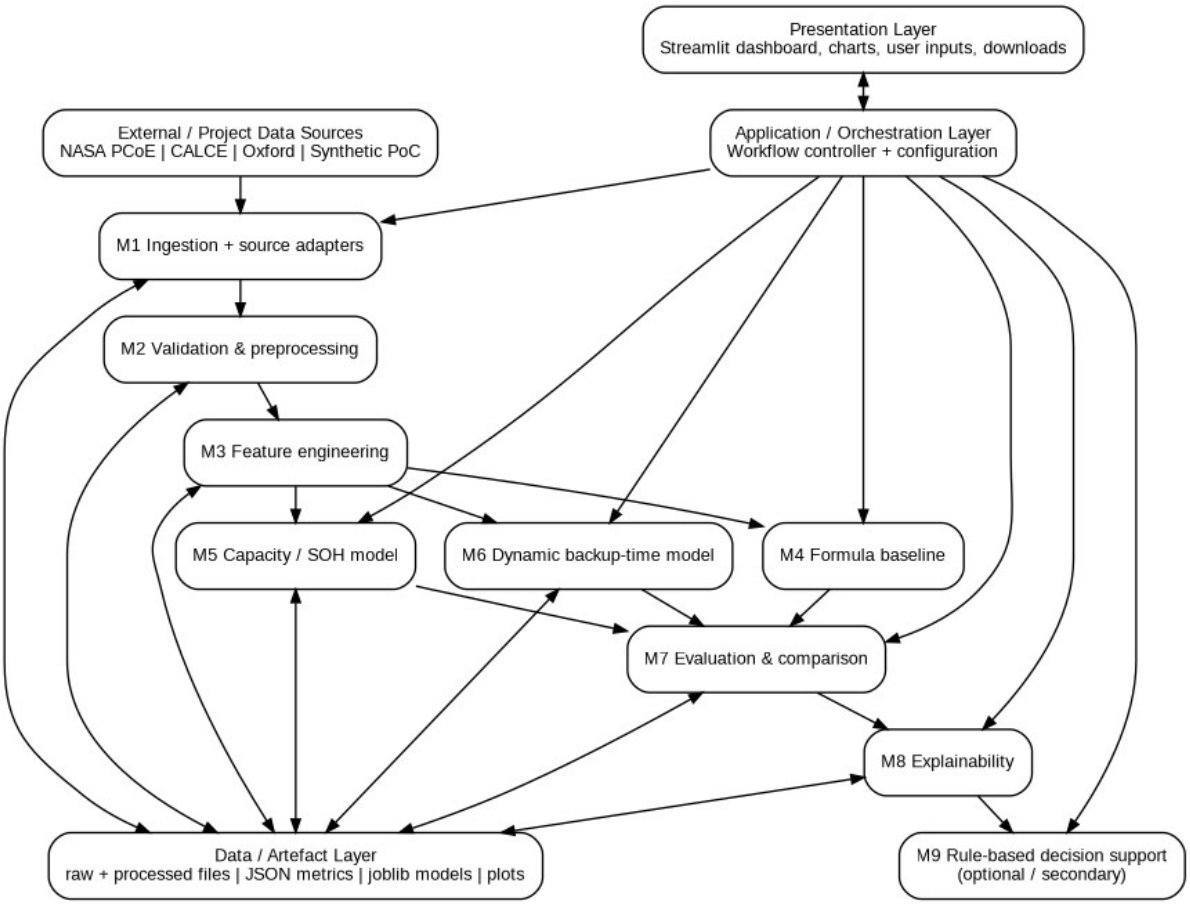


Figure 1. Layered system architecture for the proposed framework.

Figure 1 separates data sources, application services, presentation, and persisted artefacts. The dashboard communicates with an orchestration layer rather than directly manipulating models. M1-M3 establish a common validated feature layer; M4-M6 produce baseline and predictive outputs; M7 evaluates them; M8 explains model behaviour; M9 is an optional decision-support enhancement; and M10 presents results and exports. The artefact layer stores reproducible intermediate and final outputs.

5.2 Module-wise Design

Module	Responsibility	Input	Output	Consumes / Feeds
M1 Dataset Ingestion	Read CSV/Excel; identify source; apply column aliases; attach source/cell metadata.	File + mapping config	Standardized raw table	M2
M2 Validation & Preprocessing	Validate schema/types/units/ranges; report missing/duplicate/invalid rows; apply approved cleaning/imputation.	Standardized raw table	Clean table + quality report	M3
M3 Feature Engineering	Create leakage-safe cycle, electrical, thermal, load, energy, resistance, and degradation features.	Clean table	Capacity and runtime feature matrices	M4-M6
M4 Formula Baseline	Compute transparent runtime from predicted/available capacity, representative voltage, load power, and efficiency.	Feature row + predicted capacity	Formula runtime	M7
M5 Capacity/SOH Assessment	Train/load capacity regressor; predict remaining capacity; calculate SOH = capacity/reference × 100.	Health feature matrix	Capacity + SOH	M4, M7, M9, M10
M6 Dynamic Backup-Time ML	Train/load separate regression model for remaining backup time.	Runtime feature matrix	ML backup time	M7, M9, M10
M7 Evaluation & Comparison	Evaluate capacity and runtime models; compare formula and ML runtime on identical records.	Targets + predictions	Metrics, residuals, comparisons	M8, M10
M8 Explainability	Produce global/local attribution for trained models; store caveats.	Model + evaluation/features	Built-in/permutation importance tables and graphs; optional SHAP interface	M10
M9 Rule-Based Decision Support	Optional rules using SOH, backup time, load, and data-quality flags; no safety authority.	Predictions + thresholds	Guidance message	M10
M10 Dashboard/Reporting	Provide upload, validation feedback, model results, plots, assumptions, and downloads.	Outputs from M1-M9	Interactive UI + export files	User

5.3 Data Flow Design

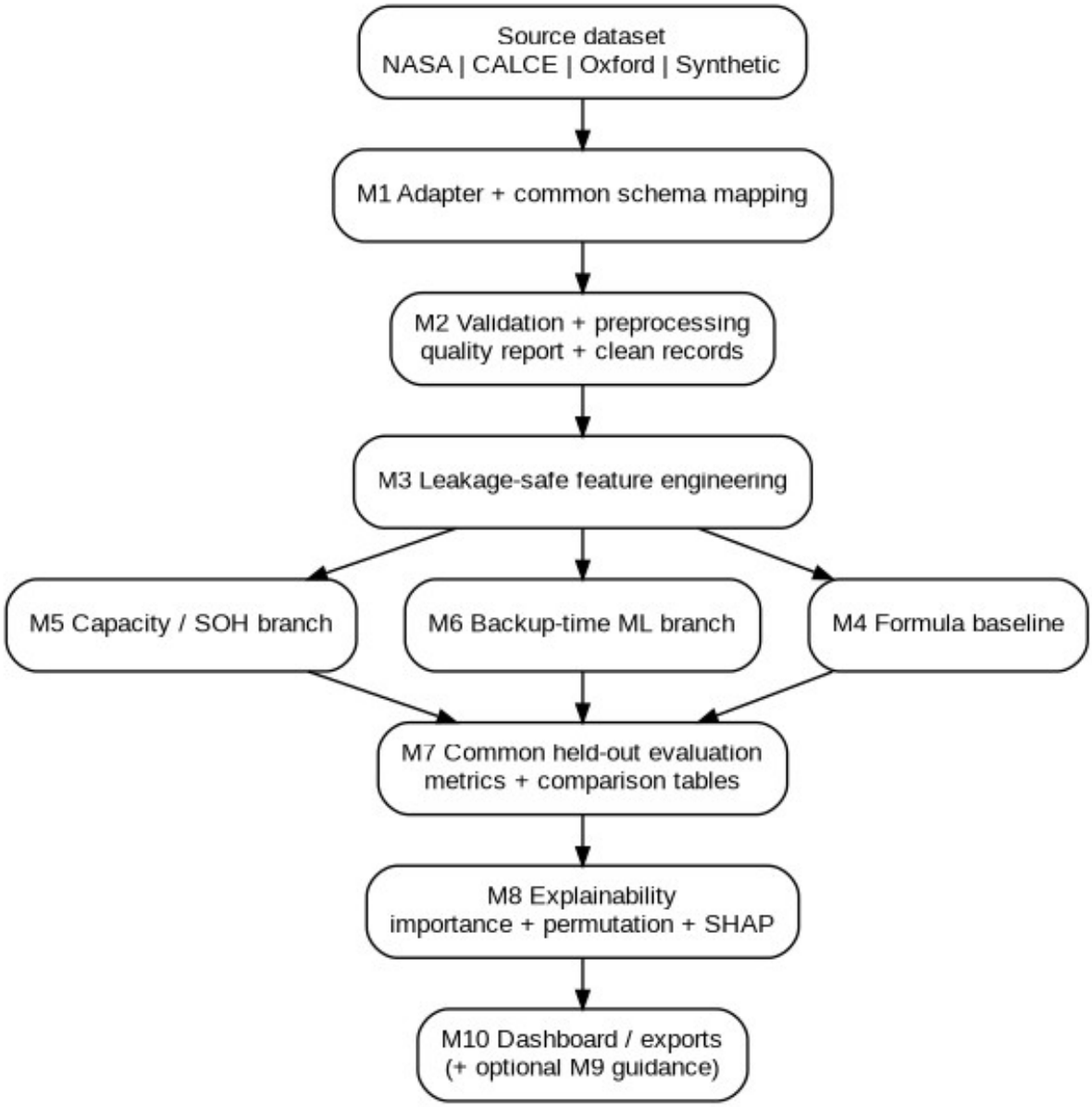


Figure 2. High-level data flow from source dataset to predictions, explanations, and exports.

1. Source identification: the user selects a supported public dataset file or project-generated test file.
2. Adapter stage: source-specific field names and metadata are mapped into the common schema without inventing unavailable measurements.
3. Validation stage: required fields and units are checked; unusable records are rejected or flagged; approved cleaning rules are applied.
4. Target construction: for datasets with discharge time series, backup time is derived only when the observation-to-cut-off definition can be reproduced without using future information as a model input.
5. Feature generation: separate capacity and runtime feature matrices are created so each target has an explicit leakage-safe feature list.
6. Prediction branches: M5 estimates capacity/SOH; M6 estimates dynamic backup time; M4 uses predicted/available capacity to calculate the transparent formula baseline.
7. Evaluation and explanation: predictions are joined to the same held-out records, scored, plotted, and explained.
8. Presentation/export: the dashboard displays data-quality information, predictions, comparisons, caveats, and downloadable files.

5.4 Data Storage / Database Design

A relational database is not required for the Phase-2 local PoC because the selected datasets are file-based and the primary aim is reproducible model development. The implementation therefore uses a file-oriented artefact design. This is consistent with the Phase-1 maintainability objective and avoids adding database complexity before the data schema and model pipeline are validated. A database may be added later without changing the logical entities below.

Location	Purpose	Representative Format
data/raw/	Original external dataset copies or local test files; read-only where possible.	CSV/Excel/source-native
data/interim/	Source-adapted and validated records plus quality reports.	CSV/Parquet + JSON
data/processed/	Model-ready feature matrices and target tables.	CSV/Parquet
models/	Capacity model, backup-time model, preprocessing objects, and metadata.	joblib + JSON
results/	Predictions, metrics, residual tables, importance values, and experiment summaries.	CSV + JSON
figures/	Actual-vs-predicted, residual, importance, and SHAP images.	PNG/SVG
config/	Column aliases, feature lists, thresholds, efficiency assumptions, random seed, and paths.	YAML/JSON
app/	Streamlit entry point and UI/service integration code.	Python

5.5 Common Data Schema

Field	Type	Model Role	Definition / Rule
record_id	string	Required internal ID	Unique observation identifier created during adaptation.
source_dataset	string	Required metadata	Identifies NASA/CALCE/Oxford/synthetic source.
cell_id	string	Required when available	Battery/cell identifier used to avoid accidental mixing and support grouped evaluation.
cycle_count	integer	Core feature when available	Cycle/aging indicator; mapping is source-specific.
timestamp_s	float	Conditional	Time within discharge; required for derived remaining-time target when applicable.
voltage_v	float	Core measurement	Measured terminal voltage; may be summarized into cycle features.
current_a	float	Core measurement	Measured charge/discharge current with documented sign convention.
temperature_c	float	Optional/core where available	Measured battery or chamber temperature.
internal_resistance_ohm	float	Optional feature	Direct or derived only when source supports a defensible value.
rated_capacity_ah	float	Required for SOH	Reference capacity used in SOH denominator.
measured_capacity_ah	float	Target/label	Ground-truth capacity where supplied/derived by source protocol; excluded from capacity-model inputs.
load_power_w	float	Derived/core runtime feature	Derived from voltage × current or mapped from source when directly available.
backup_time_min	float	Runtime target	Measured/derived remaining discharge time to defined cut-off; never used as runtime-model input.
chemistry	string	Metadata	Chemistry label used for reporting and domain limitation.
cutoff_voltage_v	float	Metadata/target definition	Discharge termination criterion where known.

5.6 Model and Prediction Design

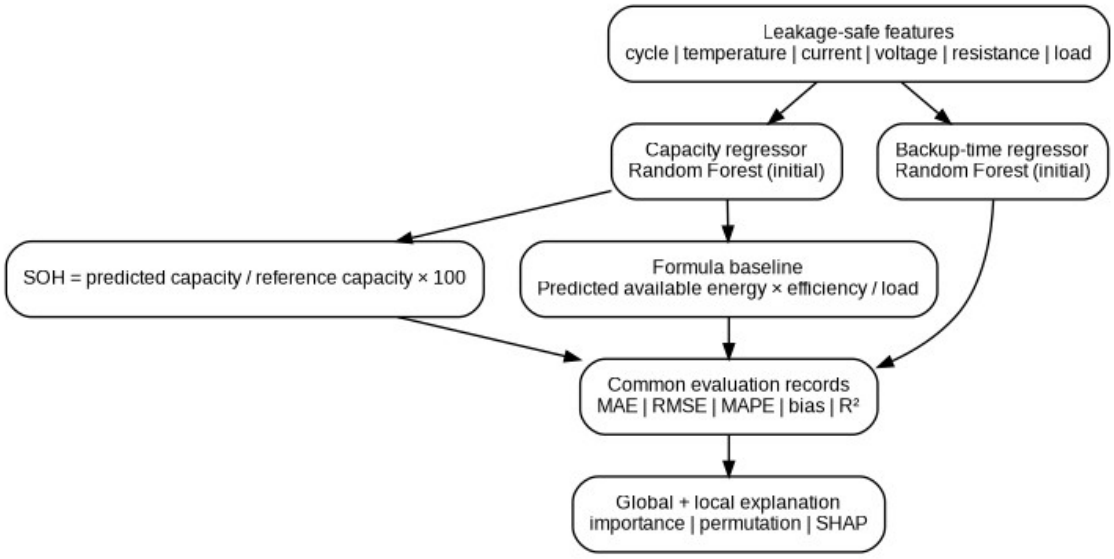


Figure 3. Separate leakage-safe health and backup-time prediction branches.

5.6.1 Capacity and SOH Branch

The initial health model is a Random Forest regressor. Its target is remaining/measured capacity, and its inputs are restricted to variables available before the target is known, such as cycle history, temperature, current/voltage summaries, and defensible resistance/degradation features. Capacity-derived labels or future-discharge information are excluded from the input feature set. SOH is then calculated as:

SOH (%) = Predicted Capacity (Ah) / Reference Capacity (Ah) × 100

5.6.2 Dynamic Backup-Time Branch

A separate Random Forest regressor predicts remaining backup time. The feature list can include current/load, voltage, temperature, cycle information, resistance/degradation indicators, and other values available at the prediction point. A dataset-specific target-construction function defines the remaining time to a documented discharge cut-off. Any future samples used to compute the target are excluded from model inputs at that observation point.

5.6.3 Formula Baseline

The transparent baseline uses the available-energy relationship defined in Phase 1. At inference, the formula should use predicted or otherwise available capacity rather than the hidden ground-truth capacity label. In simplified form:

Backup Time (min) = [Predicted Capacity (Ah) × Representative Voltage (V) × Efficiency / Load Power (W)] × 60

Efficiency remains configurable and the formula is treated as a benchmark, not a high-fidelity electrochemical model. The ML backup-time model is evaluated against the same test records so the comparison is meaningful.

6. Technology Stack and Justification

Layer / Purpose	Technology	Justification
Programming / backend	Python	Mature scientific ecosystem; clear modular code; portable across student laptops.
Data processing	pandas, NumPy	Tabular/time-series transformation, numeric operations, missing-value handling, feature generation.
Machine learning	scikit-learn	Random Forest regression, train/test utilities, evaluation metrics, permutation importance, pipelines.
Explainability	SHAP + scikit-learn permutation importance	Local/global explanation for tree models; complementary importance views.
Visualization	Matplotlib	Reproducible static model/evaluation plots that can also be displayed in Streamlit.
Frontend	Streamlit	Rapid Python-native dashboard development without a separate JavaScript frontend; suitable for a capstone prototype.
Model persistence	Joblib	Practical serialization for scikit-learn estimators and preprocessing objects.
Data / result storage	CSV/Excel, optional Parquet, JSON	Human-readable inputs/outputs and structured experiment metadata; simple local deployment.
Version control	Git + GitHub	Collaboration, issue tracking, change history, branch-based development, and reproducible releases.
Testing	pytest (planned) + manual UI tests	Automated module tests plus integration and dashboard interaction checks.
Development environment	Python virtual environment + requirements.txt	Dependency isolation and repeatable setup on team laptops.

The Phase-2 design deliberately avoids a GPU requirement and avoids introducing a web server, external API, or production database before the core battery-data and model pipeline is validated. These components can be introduced in future extensions if the project scope requires them.

7. Proof of Concept and Baseline Comparison

7.1 Baseline Data Used

The corrected Phase-2 evaluation uses the Controlled Synthetic Golden Benchmark (CSGB) v1.0 supplied in Battery_Base_Data_Benchmark_v1.0.xlsx. The workbook contains 720 deterministic observation rows, 12 complete battery groups, three controlled reference profiles, and 30 documented fields. It is designed to test ingestion, deterministic feature/formula calculations, leakage controls, model wiring, grouped evaluation, rule logic, and reporting.

Provenance boundary: The rows are synthetic analogues of selected public-domain battery-use patterns; they are not copied measurements from NASA PCoE, Oxford, CALCE, or second-life field trials (Saha & Goebel, 2007; Howey & Birkl, 2017; CALCE Battery Research Group, n.d.). Consequently, the results below are controlled functional/model-sanity evidence, not real-world accuracy evidence.

The utilized input range covers cycle number 0–1000, ambient temperature 15–35 °C, discharge current 0.25–4.00 A, C-rate 0.20–2.03 h^{-1}, internal resistance 0.02500–0.08575 Ω, measured average voltage 3.370–3.689 V, and load power 0.874–14.455 W. Controlled reference targets span 0.4514–4.8100 Ah capacity, 61.0–101.2% SOH, and 11.142–238.697 minutes actual backup time.

Baseline item	Utilized value
Workbook / sheet	Battery_Base_Data_Benchmark_v1.0.xlsx / Benchmark Data
Version and purpose	CSGB_v1.0; deterministic functional and model-sanity benchmark
Coverage	720 rows, 12 battery groups, 3 profiles, 30 documented fields
Development data	540 rows from 9 complete batteries (180 per profile)
Holdout data	180 rows from 3 unseen batteries (60 per profile)

Baseline item	Utilized value
Split integrity	0 battery_id/lobo_group overlap between development and holdout
Baseline assumptions	efficiency_fraction = 0.90; usable_fraction = 0.95
Evidence outputs	29 primary/supplementary comparison evidence files: 11 data/report outputs and 18 PNGs (11 distinct analytical views). Four integration outputs and one validation log bring the supplied generated-evidence total to 34. The 180 x 8 project_outputs_sample.csv is an intermediate audit input and is not counted.

7.1.1 Profile Coverage

Profile	Controlled analogue	Chemistry / form	Rated capacity	Cycle range	Dev. / holdout
P_NMC5	Second-life grid-storage synthetic analogue	NMC/graphite-silicon; 21700	5.00 Ah	0-900	180 / 60
P_NASA2	NASA PCoE/random-walk synthetic analogue	Li-ion; 18650	2.00 Ah	0-800	180 / 60
P_OXF074	Oxford dynamic-load synthetic analogue	Li-ion; 740 mAh pouch	0.74 Ah	0-1000	180 / 60

7.1.2 Fields Actually Utilized

Identifiers and metadata: record_id, profile_id, source_analogue, battery_id, lobo_group, chemistry, form_factor, suggested_holdout, and data_version.

Inputs and engineered observables: cycle_number, nominal/rated/cut-off values, ambient_temp_c, discharge_current_a, C-rate, internal_resistance_ohm, voltage_sag_v, measured_avg_voltage_v, and load_power_w. The Phase-2 adapter creates cycle_count, temperature_c, avg_voltage_v, min_voltage_v, measured_capacity_ah, and backup_time_min aliases required by M1-M6.

Reference-only fields: actual_capacity_ah, actual_soh_pct, formula_baseline_min, actual_backup_time_min, health_class, required_runtime_min, expected_recommendation, and data_quality_flag. These fields are used as labels, deterministic checks, or reports and are excluded from learned feature matrices. Within CSGB only, the actual_* prefix means controlled benchmark reference target; it does not mean laboratory-measured or field-observed ground truth.

7.2 Evaluation Protocol and Leakage Controls

1. M1 loads the Benchmark Data sheet and maps benchmark names to the common Phase-2 schema while preserving record_id and battery grouping metadata.
2. M2 validates required columns, types and physical ranges. All 720 rows are accepted; no record is silently lost or imputed for this benchmark run.
3. M3 generates leakage-safe C-rate, thermal stress, resistance-voltage-drop and voltage-sag features without using capacity, SOH, runtime, class or recommendation targets as inputs.
4. The predefined battery-group split trains on 540 DEVELOPMENT rows from nine complete batteries and tests on 180 HOLDOUT rows from three unseen batteries. The verified group-overlap count is zero. The 180 cycle-level rows must not be interpreted as 180 statistically independent battery units.
5. M5 produces grouped out-of-fold SOH predictions inside the development set before M6 training. M5 is then fitted on development data and predicts capacity/SOH for the holdout batteries.
6. M4 and M6 are evaluated on the identical 180 holdout record IDs. The operational M4 comparator receives M5-predicted capacity; only the separate known-answer arithmetic check is allowed to use actual capacity.
7. M9 converts M5/M6 predictions into health and recommendation outputs, which are compared with baseline labels. All aligned outputs are exported for row-level audit.

7.2.1 Supplementary Record-Level Comparison Using phase2_compare_and_plot.py

After the primary M1-M9 comparison run, phase2_compare_and_plot.py was executed as a second, separate record-level audit. It does not retrain M4, M5, M6 or M9 and does not introduce additional benchmark observations. Instead, it reads the saved project outputs, filters the Benchmark Data sheet to HOLDOUT, validates the required fields, checks unique record_id values, and performs a one-to-one merge against the same 180 reference records from three unseen battery groups.

Verified project-output interface: project_outputs_sample.csv contains 180 rows and eight fields: record_id, m4_formula_check_min, m4_operational_formula_min, m5_capacity_output_ah, m5_soh_output_pct, m6_backup_time_output_min, m9_health_class_output, and m9_recommendation_output. These are mapped

from the corresponding M4/M5/M6/M9 columns in phase2_predictions_vs_baseline.csv; no baseline target is copied into a learned prediction field.

Alignment and validation controls: The input check confirmed shape (180, 8), zero duplicate record IDs and zero missing values. The merged evidence contains all 180 requested HOLDOUT IDs, with no many-to-one or missing-ID join, and preserves the baseline battery_id, profile_id and suggested_holdout fields for audit.

Generated output data: The comparison_results folder contains the following separately regenerated post-processing evidence:

- aligned_record_comparison.csv - 180 rows and 23 columns containing baseline references, project outputs, signed errors, and M9 match flags.
- comparison_metrics_table.csv - 27 report-ready metric rows covering five regression comparisons and two exact-match classification comparisons.
- comparison_metrics.json - record count, three compared battery groups, split/join metadata, complete metrics, formula tolerance result, and the controlled-synthetic warning.
- graphs/ - seven PNG figures covering M4, M5, M6 and M9 comparisons.

Supplementary graph	Comparison	Verified interpretation
01_m4_formula_comparison.png	M4 formula check	Identity-line agreement; maximum absolute difference 5.68 × 10^-14 min after CSV round-trip, below the 0.01-min criterion; pass.
02_m5_capacity_comparison.png	M5 capacity	MAE 0.056 Ah, RMSE 0.105 Ah, MAPE 3.29%, and R^2 0.995; most points remain close to the identity line.
03_m5_soh_comparison.png	M5 SOH	MAE 2.600 percentage points, RMSE 6.204 percentage points, MAPE 3.29%, and R^2 0.657; cross-profile outliers remain visible.
04_m4_m6_runtime_comparison.png	M4 and M6 runtime	Both methods use the same 180 actual_backup_time_min records; M6 points are materially closer to the identity line than M4.
05_runtime_error_metrics.png	Runtime errors	M6 MAE/RMSE/MAPE are 1.519 min, 2.269 min and 2.91%, versus 8.756 min, 10.806 min and 18.45% for M4.
06_representative_record_comparison.png	Representative records	Actual, M4 and M6 runtime bars are shown for 12 range-spanning holdout IDs, retaining individual deviations for audit.
07_m9_accuracy_comparison.png	M9 labels	Predictive exact-match accuracy is 86.11% for health class and 88.33% for recommendation.

The runtime-error graph confirms that M6 reduces MAE by 82.66%, RMSE by 79.01%, and MAPE by 84.25% relative to the operational M4 formula on the controlled holdout set. The seven supplementary plots intentionally overlap with the primary comparison set: their purpose is a separate record-ID alignment and metric-recomputation audit, not an independent model experiment or independent validation dataset.

As with the primary comparison, these results apply only to CSGB v1.0. They demonstrate reproducible joining, metric calculation, reporting and graph generation, but do not establish real-world battery accuracy, operational fitness or safety performance.

7.3 Project Output-to-Baseline Mapping

Module	Project output	Baseline reference	Comparison method
M4	m4_reference_formula_min	formula_baseline_min	Maximum absolute difference; pass if ≤ 0.01 min
M4/M7	m4_operational_formula_min	actual_backup_time_min	MAE, RMSE, MAPE, bias, R^2
M5	m5_predicted_capacity_ah	actual_capacity_ah	MAE, RMSE, MAPE, bias, R^2
M5	m5_predicted_soh_pct	actual_soh_pct	MAE/RMSE in percentage points; MAPE, bias, R^2
M6	m6_predicted_backup_time_min	actual_backup_time_min	MAE, RMSE, MAPE, bias, R^2
M9	m9_predicted_health_class	health_class	Exact-match accuracy
M9	m9_predicted_recommendation	expected_recommendation	Exact-match accuracy

M4 fairness rule: The exact calculation check and the operational predictor are intentionally separate. Reproducing formula_baseline_min verifies M4 arithmetic using the known reference capacity; comparing runtime performance uses M5-predicted capacity so M4 does not receive a hidden target unavailable to M6.

7.4 Quantitative Results

Output	Reference	MAE	RMSE	MAPE	Bias	R^2
M4 formula check	formula_baseline_min	~0.000 min	~0.000 min	~0.00%	~0.000 min	1.000
M5 capacity	actual_capacity_ah	0.056 Ah	0.105 Ah	3.29%	+0.037 Ah	0.995
M5 SOH	actual_soh_pct	2.600 pp	6.204 pp	3.29%	+2.212 pp	0.657
M4 operational runtime	actual_backup_time_min	8.756 min	10.806 min	18.45%	+8.466 min	0.944
M6 ML runtime	actual_backup_time_min	1.519 min	2.269 min	2.91%	+0.868 min	0.998

M4 reproduced the workbook formula baseline with a maximum absolute difference of 2.84×10^{-14} minute, passing the proposed ≤ 0.01-minute arithmetic criterion. On the fair operational comparison, M6 reduced MAE by 82.66%, RMSE by 79.01%, and MAPE by 84.25% relative to M4 on the same holdout IDs.

M9 predictive outputs achieved 86.11% exact health-class agreement and 88.33% exact recommendation agreement. Macro F1 is 81.39% and 82.00%, respectively. The most consequential minority labels - Replace Planning and Plan replacement - each have 50.00% recall (7 correct of 14 reference cases), so aggregate accuracy must not be used alone. When baseline reference SOH/runtime values were supplied solely to test deterministic rule wiring, both rule checks achieved 100%; this reference-input unit test is not a predictive-accuracy claim.

Optional row-level screens were also reported: 93.89% of capacity predictions were within 5% absolute percentage error, 90.56% of SOH predictions were within 3 percentage points, and 97.78% of M6 runtime predictions were within 10% absolute percentage error.

M8 global explanation evidence is exported for both learned models. Built-in and 10-repeat permutation importance identify internal_resistance_ohm as the leading M5 feature and c_rate as the leading M6 feature on CSGB. Because the benchmark features are correlated and synthetically constructed, these rankings describe model behaviour and are not causal battery-physics conclusions.

7.4.1 Runtime Results by Profile

Profile	Rows	M4 MAE (min)	M4 MAPE	M6 MAE (min)	M6 MAPE	M6 R^2
P_NASA2	60	8.009	20.25%	1.054	2.64%	0.999
P_NMC5	60	9.691	11.25%	2.249	2.71%	0.995
P_OXF074	60	8.567	23.85%	1.252	3.37%	0.997

7.5 Output and Comparison Graphs

Figures 4-11 are generated from phase2_predictions_vs_baseline.csv. Figures 12-13 use the fitted M5/M6 models and controlled holdout features for global importance, while Figure 14 uses the aligned M9 reference/predicted labels. Each runtime method uses the same 180 grouped holdout rows. The phase2_compare_and_plot.py plots are a separate record-level recomputation audit, not independent model validation.

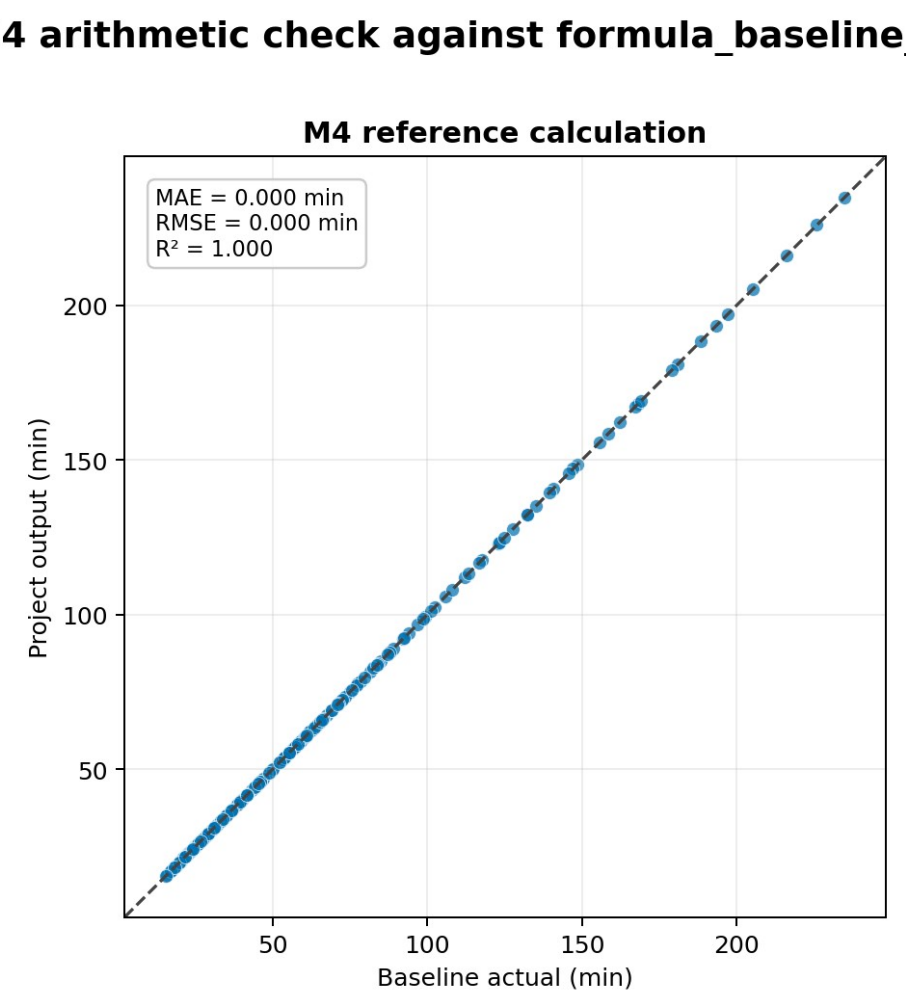


Figure 4. M4 project formula output versus formula_baseline_min. The exact arithmetic check passes the ≤ 0.01-minute criterion.

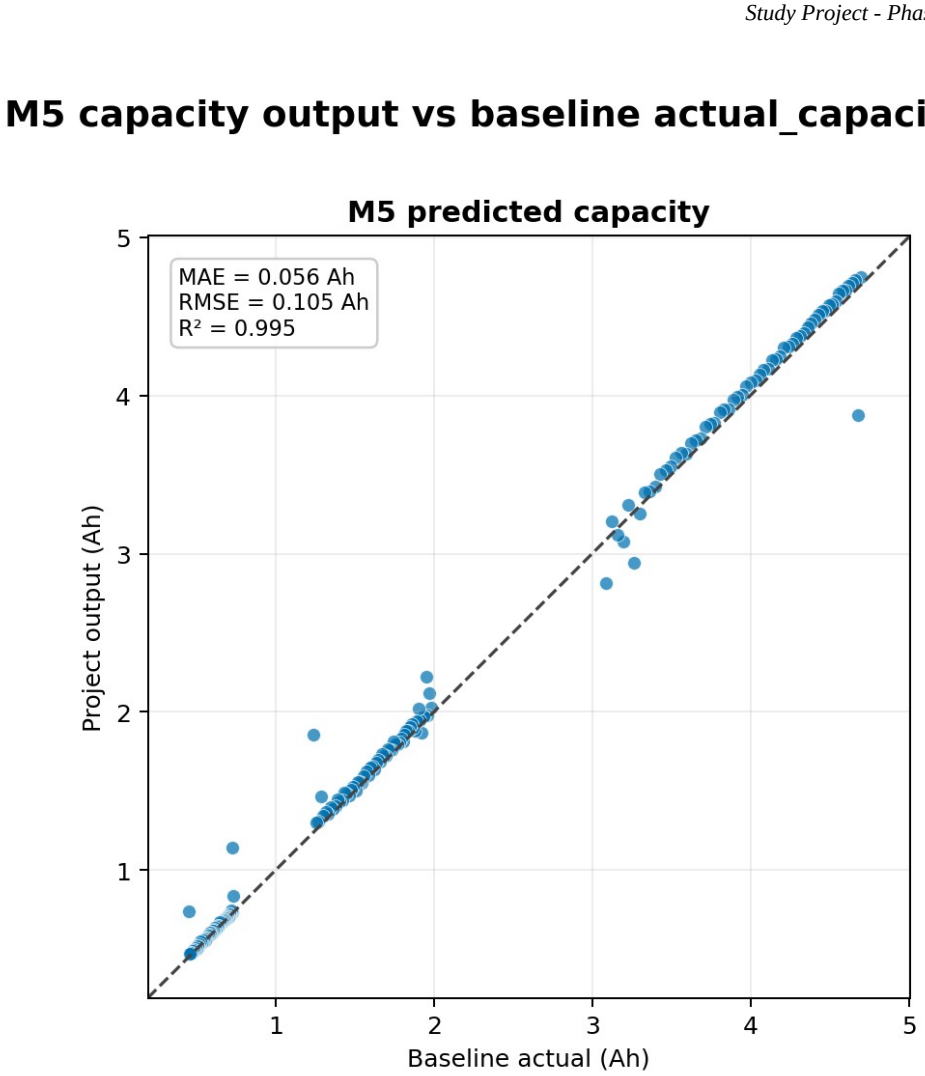


Figure 5. M5 predicted capacity versus actual_capacity_ah for the 180 unseen-battery holdout records.

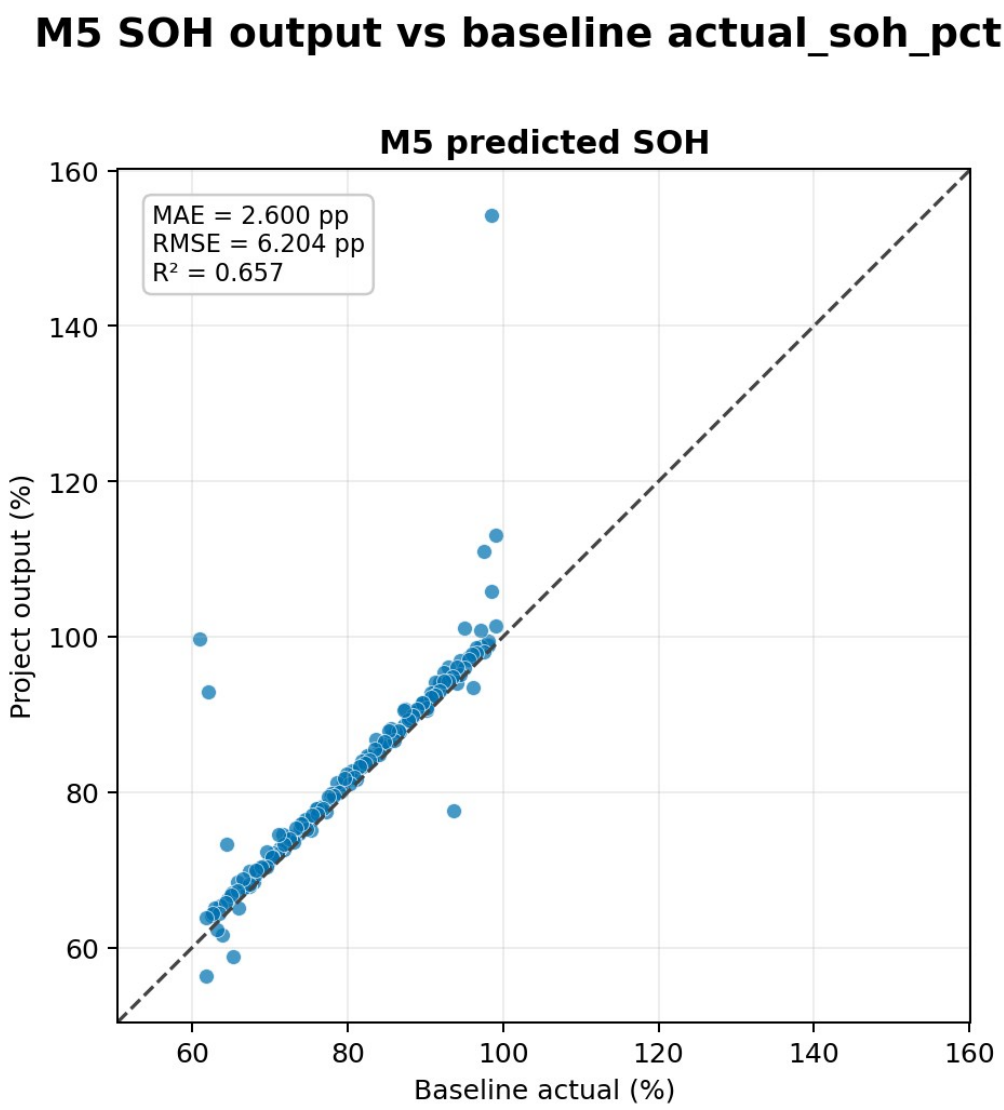


Figure 6. M5 predicted SOH versus actual_soh_pct. Cross-profile outliers explain the lower SOH R^2 relative to capacity R^2.

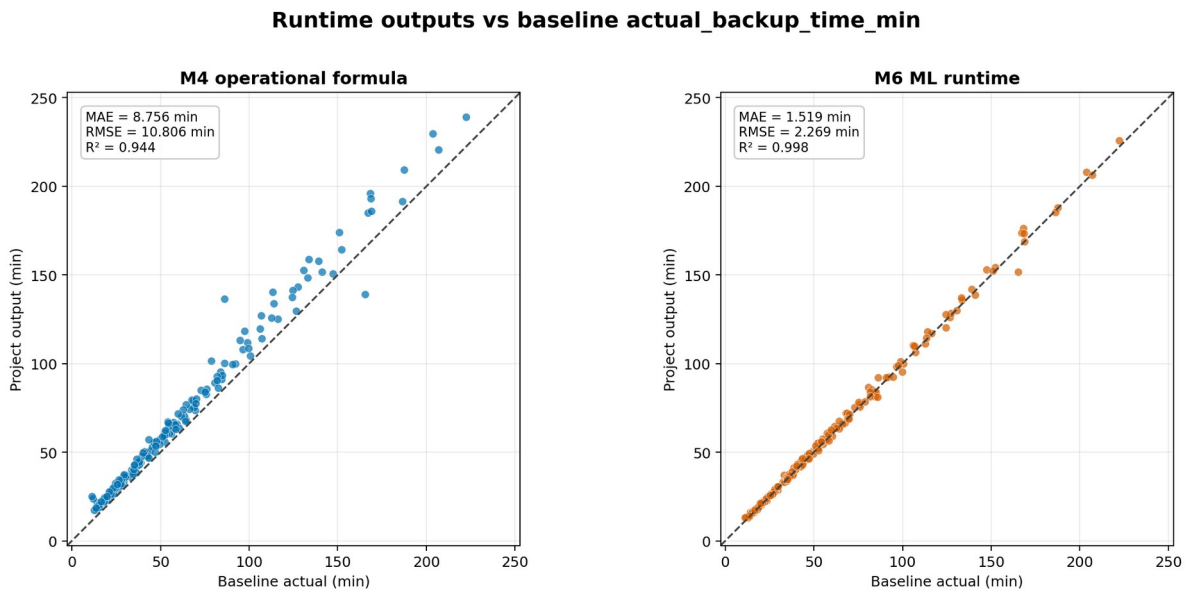


Figure 7. M4 operational formula and M6 ML runtime outputs versus actual_backup_time_min on identical holdout records.

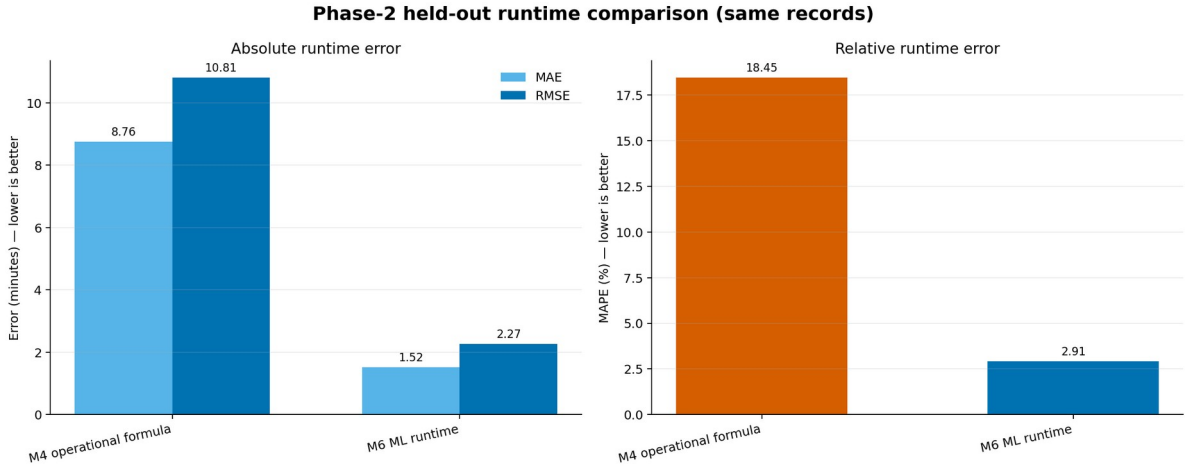


Figure 8. Runtime error comparison. M6 has substantially lower MAE, RMSE and MAPE than the operational M4 formula on this controlled benchmark.

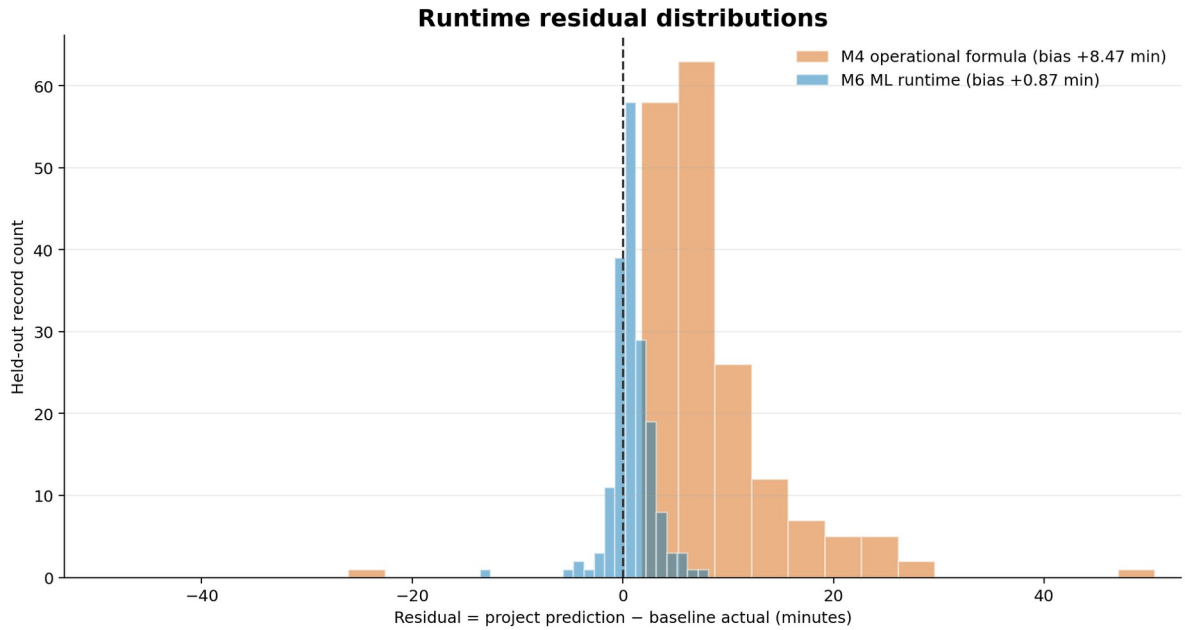


Figure 9. Runtime residual distributions, where residual equals prediction minus actual. M6 residuals cluster more closely around zero.

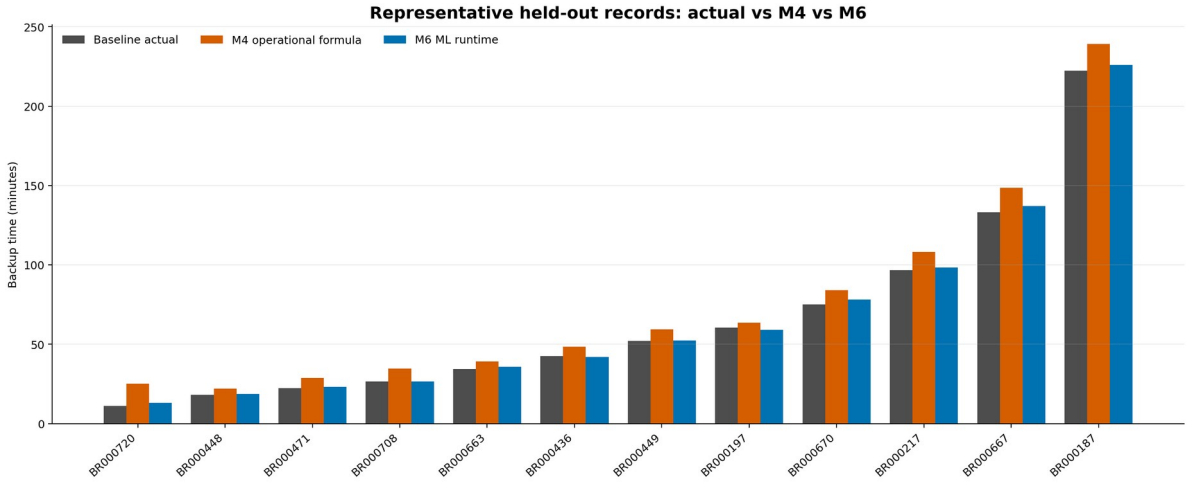


Figure 10. Representative holdout record comparison of baseline actual runtime, M4 operational formula and M6 ML runtime.

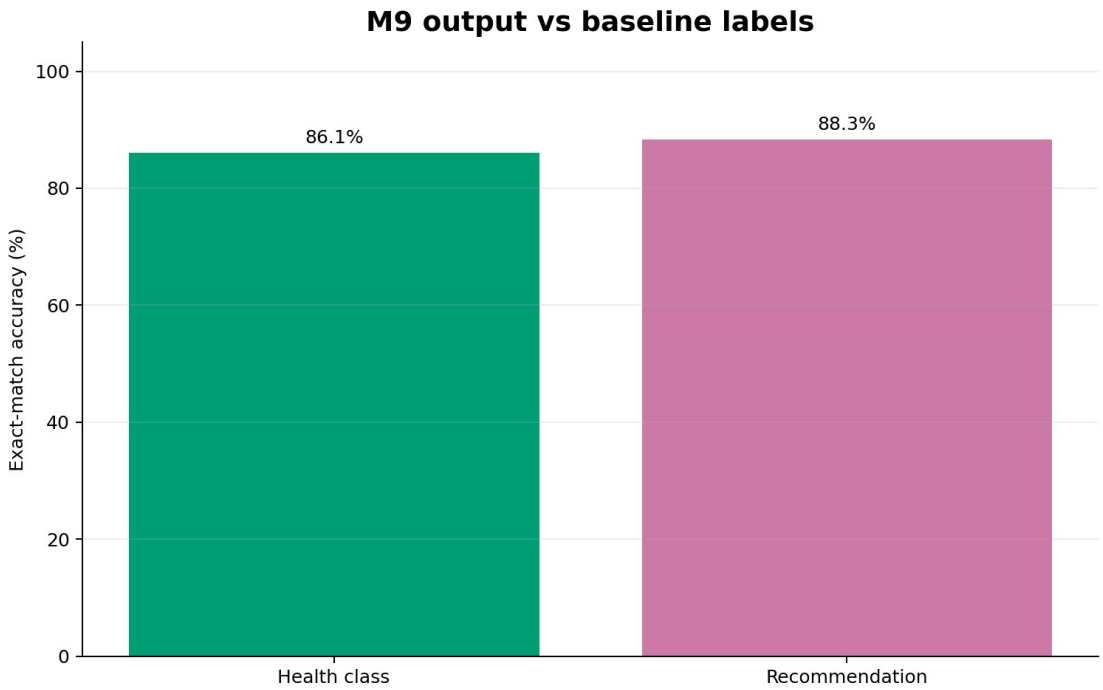


Figure 11. M9 predictive health-class and recommendation exact-match accuracy against baseline labels.

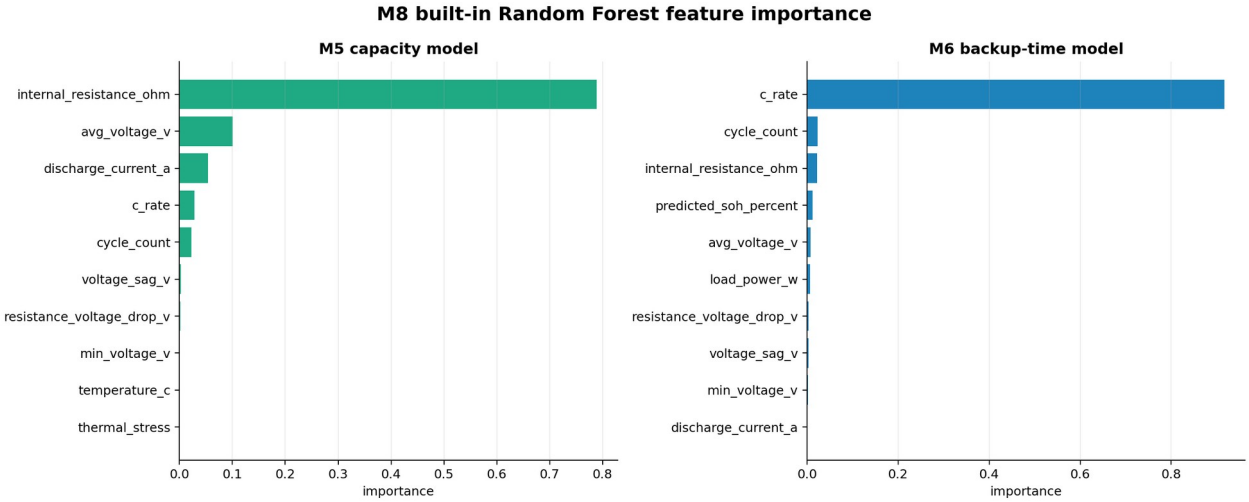


Figure 12. M8 built-in Random Forest feature importance for M5 capacity and M6 backup-time models. Values describe the fitted controlled-benchmark models and are not causal effects.

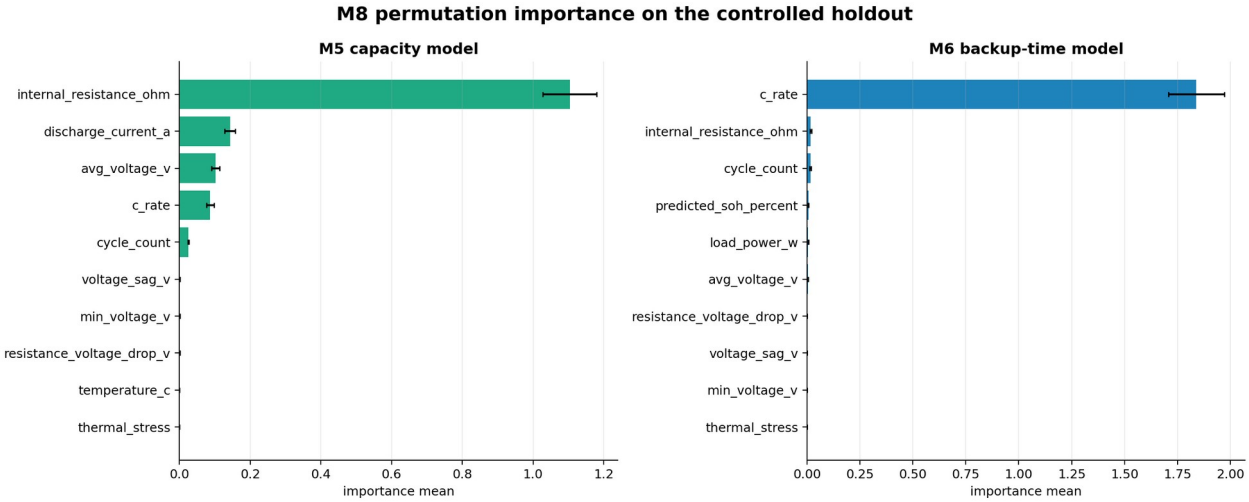


Figure 13. M8 ten-repeat permutation importance on the controlled holdout. Error bars show repeat-to-repeat standard deviation; correlated inputs can redistribute importance.

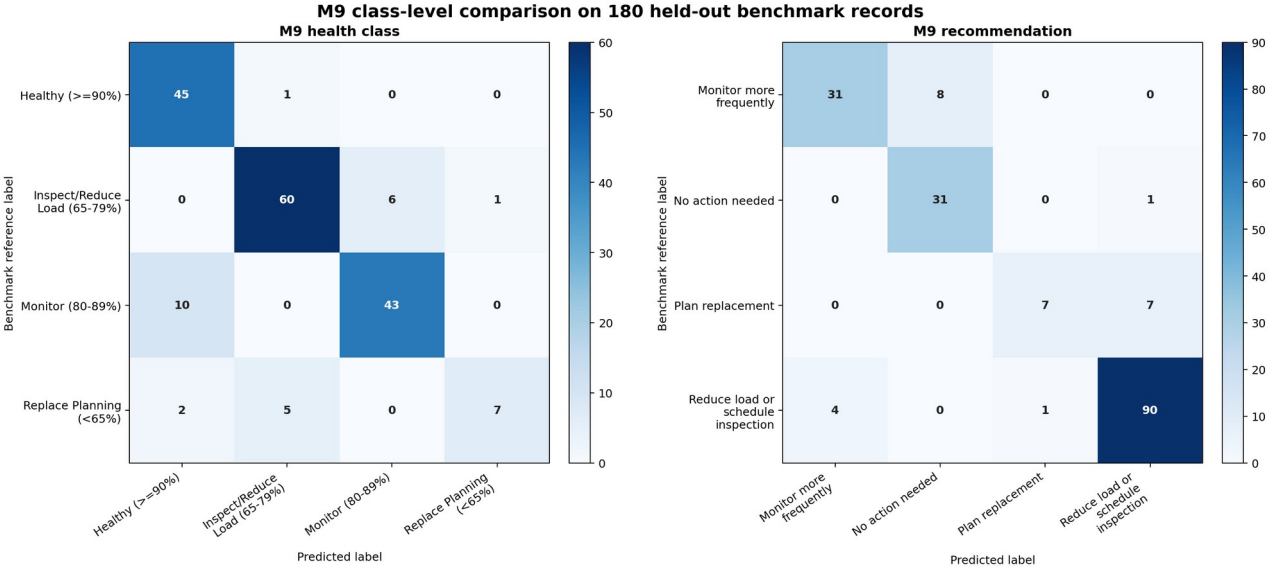


Figure 14. M9 health-class and recommendation confusion matrices. The replacement-related reference classes each achieve 50% recall, showing why exact-match accuracy must be supplemented by class metrics.

7.6 Findings and Limitations

- The formula implementation is numerically correct against the controlled reference and the group split prevents battery-level overlap between development and holdout data.
- M5 capacity performance is strong on the benchmark, but the lower SOH R^2 and visible cross-profile outliers show that capacity-to-SOH transfer across different rated capacities requires further calibration and real-data testing.
- M6 materially outperforms the simplified operational formula on this controlled dataset; the result demonstrates model-path feasibility rather than universal superiority.
- M8 built-in and permutation importance are now reproducibly exported and graphed for M5/M6. Their agreement on the leading features supports model-path interpretability, but correlated synthetic inputs prevent causal interpretation and local SHAP remains Phase 3.
- M9 reference-input rule wiring is deterministic, whereas predictive class/recommendation accuracy depends on M5/M6 error and decision thresholds. Class-level analysis shows 50% recall for both replacement-related labels, so threshold refinement and real-data validation remain necessary despite higher aggregate accuracy.
- CSGB does not reproduce every electrochemical, sensor, protocol, or time-series effect present in experimental batteries. Phase 3 must repeat grouped evaluation on compatible NASA PCoE, CALCE, and Oxford experimental cells and report results by dataset/protocol (Saha & Goebel, 2007; CALCE Battery Research Group, n.d.; Howey & Birkl, 2017).

- No result in this section is a certified BMS decision, inverter-control limit, replacement standard, field-telemetry validation, or safety recommendation.

7.7 Consolidated Phase-2 Output Inventory and Capstone Achievement Assessment

This subsection consolidates the evidence created by the primary M1-M9 baseline run, the separate phase2_compare_and_plot.py record-level audit, the integration demonstration, and the automated validation suite. Repeated visualizations are not treated as separate experiments.

7.7.1 Exact Generated Output Count

Execution	Data/report files	PNG graphs	Total
run_phase2_baseline_comparison.py	8	11	19
phase2_compare_and_plot.py	3	7	10
Comparison evidence subtotal	11	18	29
run_phase2_m1_to_m9.py integration outputs	4	0	4
Complete validation-suite log	1	0	1
Supplied generated evidence total	16	18	34

Counting rule: The 180 x 8 project_outputs_sample.csv hand-off file is an intermediate input and is not counted. The supplied package contains 29 primary/supplementary comparison evidence files (11 data/report and 18 PNG), four additional M1-M9 integration outputs, and one complete validation log: 34 generated evidence files in total. The 18 PNG files represent 11 distinct analytical views because seven supplementary plots intentionally recompute existing views.

7.7.2 Output Data Files and Their Purpose

File / role	Dimensions	Purpose
phase2_predictions_vs_baseline.csv	180 × 42	Primary record-level evidence: identifiers, actuals, M4-M6 outputs, signed/absolute/percentage errors, screens, M9 outputs and match flags.
phase2_metrics_table.csv	27 × 7	Report-ready MAE, RMSE, MAPE, bias and R^2 rows for numeric comparisons plus two M9 exact-match results.
phase2_comparison_metrics.json	Structured JSON	Data-quality, split-integrity, formula, model, M9, optional screen and per-profile evidence for reproducibility and integration.
comparison_summary.txt	Text summary	Concise human-readable execution result and controlled-synthetic interpretation boundary.
project_outputs_sample.csv	180 × 8	Intermediate record_id plus seven M4/M5/M6/M9 outputs used by the separate post-processing audit; input file, not counted as generated evidence.
aligned_record_comparison.csv	180 × 23	One-to-one HOLDOUT merge of baseline references, project outputs, signed errors and M9 match flags.
comparison_metrics_table.csv	27 × 5	Separate recomputation of the same five regression and two exact-match comparisons from the saved hand-off interface.
comparison_metrics.json	Structured JSON	Separate record count, join/split metadata, metrics, formula tolerance, and synthetic-data warning; not an independent model experiment.
m8_builtin_feature_importance.csv	22 x 3	Global fitted-model feature importance for M5 and M6.
m8_permutation_importance.csv	22 x 4	Ten-repeat holdout permutation mean and standard deviation for M5/M6.
m9_classification_report.csv	14 x 6	Per-class precision, recall, F1 and support plus macro/weighted results.
m9_confusion_counts.csv	32 x 4	Tidy expected-vs-predicted M9 class counts behind Figure 14.
phase2_outputs/	4 files	Separate 960-row integration demonstration predictions, metrics and two importance CSVs; excluded from the 29-file comparison subtotal.
TEST_EXECUTION_LOG.txt	Complete text log	Compilation, M1-M9 standalone runs, both comparisons, integration, 11 automated tests and package verification; final status PASS.

- **Validated benchmark:** 720/720 CSGB v1.0 records were retained, representing 12 battery groups; no duplicate record_id, missing numeric output, range violation, silent row loss or imputation occurred in this run.
- **Grouped evaluation:** 540 records from nine complete batteries were used for development and 180 records from three unseen batteries were held out, with zero battery-group overlap.

- **Holdout composition:** NASA2_B04, NMC5_B04 and OXF074_B04 each contribute 60 records, enabling overall and profile-level runtime assessment.

7.7.3 Exact Quantitative Outputs and Achievements

Output	MAE	RMSE	MAPE	Bias	R^2
M4 formula check	~0.000 min	~0.000 min	~0.00%	~0.000 min	1.000
M5 capacity	0.056 Ah	0.105 Ah	3.29%	+0.037 Ah	0.995
M5 SOH	2.600 pp	6.204 pp	3.29%	+2.212 pp	0.657
M4 operational runtime	8.756 min	10.806 min	18.45%	+8.466 min	0.944
M6 ML runtime	1.519 min	2.269 min	2.91%	+0.868 min	0.998

- **M4 arithmetic verification:** The maximum absolute difference is 2.84×10^{-14} minute in the primary run and 5.68×10^{-14} minute after the supplementary CSV round-trip; both are far below the 0.01-minute acceptance criterion.
- **Runtime improvement:** Relative to the fair operational M4 formula, M6 reduces MAE by 82.66%, RMSE by 79.01% and MAPE by 84.25% on identical holdout records.
- **M9 predictive outputs: Health class matches 155/180 records (86.11%) and recommendation matches 159/180 records (88.33%). Macro F1 is 81.39% and 82.00%. Replace Planning and Plan replacement each have 50.00% recall (7/14). The separate 100% reference-input result verifies rule wiring and is not predictive accuracy.**
- **Optional row screens:** Capacity APE is within 5% for 169/180 records (93.89%), SOH absolute error is within 3 percentage points for 163/180 records (90.56%), and runtime APE is within 10% for 176/180 records (97.78%).
- **Bias interpretation:** Positive runtime bias means average over-prediction. M6 bias is much smaller than M4 bias, but optimistic backup-time error must still be investigated and calibrated on real experimental batteries before operational use.

7.7.4 What Each Distinct Graph Achieves

- **01_m4_formula_reference_check.png -** confirms arithmetic identity with formula_baseline_min. It verifies implementation, not physical backup-time accuracy.
- **02_m5_capacity_actual_vs_predicted.png -** shows how closely predicted capacity follows actual_capacity_ah and exposes capacity outliers; aggregate R^2 is 0.995.
- **03_m5_soh_actual_vs_predicted.png -** shows overall SOH agreement and cross-profile outliers. The lower R^2 of 0.657 identifies SOH transfer/calibration as a priority for real-data testing.
- **04_runtime_actual_vs_predicted.png -** compares operational M4 and M6 with the same actual_backup_time_min values; M6 points remain materially closer to the identity line.
- **05_runtime_error_metrics.png -** places M4 and M6 MAE/RMSE/MAPE side by side and quantifies the M6 error reductions.
- **06_runtime_residual_distributions.png -** reveals prediction bias, error spread and extreme cases; M4 is more positively biased while M6 residuals cluster nearer zero.
- **07_representative_runtime_records.png -** retains individual deviations for 12 range-spanning HOLDOUT records instead of relying only on aggregate metrics.
- **08_m9_exact_match_accuracy.png - summarizes aggregate health-class and recommendation agreement and is interpreted together with the class-specific confusion matrices.**
- 09_m8_builtin_feature_importance.png - ranks fitted Random Forest input features for M5 and M6; it provides a reproducible global model view.
- 10_m8_permutation_importance.png - measures the holdout R2 reduction after repeated feature shuffling and exposes ranking variability.
- 11_m9_confusion_matrices.png - shows which health/recommendation labels are confused and reveals the 50% replacement-class recall hidden by aggregate exact-match accuracy.

The seven phase2_compare_and_plot.py graphs separately recompute primary views 1-5, 7, and 8 from the 180 x 8 hand-off interface. Their added value is record-ID alignment and independent metric recomputation in code, not independent training data or an independent model experiment. The primary residual, M8 importance, and M9 confusion views are not duplicated by the supplementary script.

7.7.5 Achievement Against the Capstone Title

Title component	Phase-2 evidence	Evidence still required
Explainable	M5/M6 built-in and ten-repeat permutation importance tables/graphs are demonstrated; explanation interfaces and non-causality caveats are defined.	Global/local SHAP case explanations and feature-stability evidence on real-data grouped folds remain required.
Machine-learning framework	M1-M9 controlled-benchmark workflow, leakage controls, evaluation and exports execute reproducibly.	Model/preprocessor persistence, reload consistency and complete Streamlit integration remain required.
Battery health assessment	M5 predicts capacity and SOH for all 180 unseen synthetic holdout records.	Grouped real-battery capacity/SOH evaluation and degradation trajectories remain required.
Dynamic backup-time prediction	M6 is compared fairly with operational M4 on identical synthetic HOLDOUT records.	Real discharge-duration target construction and validation across load, temperature and health conditions remain required.

Phase 2 therefore achieves its intended design-and-PoC target: it demonstrates a modular, tested, auditable, quantitatively evaluated, and globally explainable implementation on controlled data. It does not complete the final capstone claim because public experimental validation, local/global SHAP case explanations, persistence, and the integrated dashboard remain open Phase-3 requirements.

7.7.6 Minimum Additional Phase-3 Evidence

- **Data provenance and target construction:** document NASA/CALCE/Oxford-compatible source adapters, discharge cut-off rules, raw-to-valid cycle counts, exclusions and final target records.
- **Grouped generalization evaluation:** report per-battery, per-dataset and per-profile metrics with grouped folds and mean/dispersion or confidence intervals.
- **Health and degradation views:** add actual-versus-predicted capacity/SOH graphs and actual/predicted capacity and SOH trajectories against cycle number for held-out batteries.
- **Runtime diagnostics:** compare formula and ML runtime by dataset and inspect residuals against cycle, load power, temperature, internal resistance and SOH.
- **Explainability: add M5/M6 global SHAP summary/bar plots, local SHAP explanations for accurate/over/under-predicted cases, and feature-ranking stability across real-data folds. Built-in and permutation importance are complete for the controlled Phase-2 benchmark.**
- **Decision-support validation: repeat confusion matrices, precision, recall, F1-score, support, and threshold-sensitivity analysis on real experimental evaluation records before recalibrating guidance rules.**
- **Robustness and implementation:** add uncertainty/calibration analysis, simple-model or ablation comparisons, model reload consistency, inference timing and Streamlit end-to-end tests.

No further duplicate synthetic comparison graphs are required for the Phase-2 claim. The next figures should add real-data degradation, local explanation, uncertainty, threshold-sensitivity, and robustness evidence.

8. Testing and Validation Strategy

8.1 Unit Testing Approach

Unit / Function	Key Tests
Schema validator	Required fields; 720-row acceptance; numeric conversion; unit/range rules; duplicate and missing reporting.
Target builder	Remaining-time calculation from ordered discharge timestamps; cut-off handling; no negative times.
Feature builder	Correct units, deterministic feature names, no target columns in feature lists.
Formula baseline	Known-answer workbook comparison; vector/scalar inputs; positive load; efficiency/usable-fraction bounds.
SOH calculator	Reference equation check plus predicted capacity/rated-capacity conversion; invalid denominator handling.
Metrics	Aligned known arrays and 180-row holdout outputs with independently checkable MAE/RMSE/MAPE/bias/R^2.
Model persistence	Saved model/preprocessor reload produces same prediction within numerical tolerance.
Recommendation rules	Health/recommendation boundary values; 100% reference-input wiring check; non-safety-critical wording.

Unit / Function	Key Tests
Explainability evidence	M5/M6 built-in coverage and normalized importance; permutation coverage, finite values, exported CSVs and graphs.
Automated validation suite	Compile all scripts; execute M1-M9 individually; run primary, supplementary and integration workflows; run 11 unit/evidence tests; verify package contracts.

8.2 Integration Testing Approach

- M1 → M2: source adapter output must satisfy the validator contract.
- M2 → M3: cleaned columns and metadata must be sufficient for the selected feature builder.
- M3 → M5/M6: feature columns must match the persisted model/preprocessor contract and exclude prohibited targets.
- M5 → M4: formula baseline receives predicted/available capacity, not hidden ground-truth capacity during inference.
- M4/M5/M6 → M7: comparison table uses identical record IDs and evaluation split.
- M7/M9 → reporting: metrics, decision-support labels, and graphs must correspond to the same aligned record IDs and run.
- Export path: phase2_predictions_vs_baseline.csv, metric CSV/JSON files, summary text, and PNG graphs must contain values from the same run.

8.3 Machine-Learning Validation Strategy

- Use fixed, documented splits for reproducibility and group by cell/battery where random row-level splitting would leak battery-specific information.
- Fit preprocessing and feature-selection steps on training data only.
- Report MAE and RMSE as primary regression errors; report MAPE only when target values are sufficiently above zero; also report bias and R^2 where valid.
- Compare the formula baseline and ML backup-time prediction on the exact same held-out records.
- Report performance separately by dataset/chemistry/protocol where heterogeneous data make a single pooled metric misleading.
- Use compatible cross-dataset evaluation where possible; otherwise describe the incompatibility rather than forcing a comparison.
- Inspect residual/error distributions and extreme cases, not only average metrics.

8.4 Phase-2 Controlled Benchmark Test Cases

Test ID	Feature / Scenario	Expected Result	Status
TC-P2-01	CSGB workbook contains the documented benchmark schema and unique record IDs.	Validator accepts all 720 records and preserves 12 battery groups.	Pass
TC-P2-02	Capacity feature list is inspected for target leakage.	No actual capacity, SOH, runtime, class, recommendation or generator parameter is used as an M5 input.	Pass
TC-P2-03	Runtime feature list and M5-to-M6 hand-off are inspected.	actual_backup_time_min is absent; development SOH input is grouped out-of-fold.	Pass
TC-P2-04	Predefined grouped development/holdout split is applied.	540 development rows, 180 holdout rows and zero battery overlap.	Pass
TC-P2-05	M4 known-answer formula check is executed.	Maximum absolute difference ≤ 0.01 minute.	Pass
TC-P2-06	M5 predicts capacity and SOH for every holdout record.	180 finite capacity/SOH outputs aligned by record_id.	Pass
TC-P2-07	M6 predicts dynamic backup time for every holdout record.	180 finite runtime outputs; MAE/RMSE/MAPE/bias/R^2 produced.	Pass
TC-P2-08	M4 operational and M6 runtime are compared fairly.	Both methods scored on the same 180 actual_backup_time_min references.	Pass
TC-P2-09	Output and comparison graph generation is executed.	Eleven primary PNG graphs and seven supplementary audit graphs are saved and visually verified.	Pass
TC-P2-10	M9 health and recommendation outputs are evaluated.	Exact matches, per-class precision/recall/F1/support and confusion counts reported; reference-input rule wiring = 100%.	Pass
TC-P2-10A	Supplementary phase2_compare_and_plot.py record-level audit is executed.	180 unique HOLDOUT IDs merge one-to-one; aligned CSV, metric CSV/JSON and seven PNG graphs are generated, with summary metrics matching Section 7.4.	Pass

Test ID	Feature / Scenario	Expected Result	Status
TC-P2-10B	M8 global explanation evidence is generated for M5 and M6.	Built-in and ten-repeat permutation CSVs plus two graphs are produced with complete feature coverage and finite values.	Pass
TC-P2-10C	Complete reproducibility suite is executed.	M1-M9 run individually, both comparisons and integration complete, 11 automated tests pass, and package verifier passes.	Pass
TC-P2-11	Public experimental dataset adapter validation.	Dataset-specific mapping and target construction verified against source files.	Planned - Phase 3
TC-P2-12	Streamlit end-to-end workflow.	Upload → validation → prediction → explanation → export.	Planned - Phase 3

9. Risks, Challenges, and Mitigation

Risk	Probability	Impact	Mitigation	Phase-2 Review
Heterogeneous public datasets	High	High	Use source-specific adapters, preserve metadata, validate units, and avoid forcing unsupported common fields.	Open - primary Phase-3 risk
Target leakage	Medium	High	Separate capacity/runtime feature sets; review target construction; group splits where needed.	Controlled in CSGB run: grouped split, zero overlap and OOF M5-to-M6 SOH
Controlled benchmark metrics misinterpreted	High	High	Label CSGB as synthetic sanity evidence; separate exact formula/rule tests from predictive results; prohibit real-world accuracy claims.	Mitigated in corrected Section 7 and figure captions
Cross-chemistry generalization	High	High	Report per-dataset results and retrain/calibrate for new domains; no chemistry-agnostic accuracy claim.	Open
Formula assumptions	Medium	Medium	Keep efficiency configurable; use predicted/available capacity; document simplifications.	Controlled
Explainability distorted by correlated features	High	Medium	Use permutation + SHAP + conventional importance; include non-causality caveat.	Partly mitigated: built-in and ten-repeat permutation evidence are paired with a non-causality caveat; SHAP/local/stability evidence remains open.
Model overfitting	Medium	High	Held-out/grouped evaluation, constrained RF complexity, cross-dataset checks where valid.	Open for real data
Dashboard scope / time pressure	Medium	High	Prioritize upload, validation, predictions, metrics, explanations, and export; keep recommendation module secondary.	Controlled by scope
Dependency/environment problems	Medium	Medium	Pinned core and optional explainability requirements, Python 3.12 environment record, deterministic seeds, full validation runner and log.	Controlled for the submitted Phase-2 package
Prototype used as safety authority	Low	High	Persistent limitation message; no direct control output; require expert/field validation for operational use.	Controlled

10. Phase 2 Outcomes and Readiness for Phase 3

10.1 Phase-2 Outcomes

- The final Phase-1 title, primary objectives and scope boundary have been translated into a detailed M1-M9 analytical workflow with a proposed dashboard/reporting layer.
- A layered architecture, module contracts, data-flow model, artefact-storage layout, and common battery-data schema have been defined.
- The capacity/SOH and dynamic backup-time models are separated into leakage-safe workflows and a transparent formula baseline is integrated into the comparison design.
- Technology choices have been justified around a Python/scikit-learn/Streamlit stack that can run on a normal laptop.
- A reproducible CSGB v1.0 evaluation with 720 records has demonstrated 720/720 validation, a 540/180 battery-group split with zero overlap, M4 formula verification, M5 capacity/SOH prediction, M6 runtime

prediction, M8 built-in/permutation importance, M9 aggregate and class-level comparison, aligned exports, and 11 primary graphs.

- A separate phase2_compare_and_plot.py audit rejoined the same 180 HOLDOUT outputs by record_id and recomputed key metrics/plots. The package supplies 29 comparison evidence files, four integration outputs, and one validation log (34 generated evidence files total), including 18 PNGs representing 11 distinct analytical views. The hand-off CSV is documented separately as an input.
- Functional requirements, non-functional requirements, test strategy, 11 automated module/evidence tests, individual M1-M9 demonstrations, a complete execution log, and a revised risk register are documented.
- Controlled synthetic benchmark results are explicitly separated from empirical validation; no real-world accuracy or safety claim is made at Phase 2.

10.2 Items Carried Forward to Phase 3

- Implement and verify source adapters for the selected NASA PCoE, CALCE subset, and Oxford Battery Degradation Dataset 1.
- Implement reproducible discharge target construction for each compatible source dataset and document cut-off definitions.
- Repeat the verified grouped/held-out evaluation protocol on compatible real public experimental batteries and report fold/profile results separately.
- Extend the real-data evidence pack with held-out degradation trajectories, residuals stratified by cycle/load/temperature/SOH, repeated M9 class diagnostics, uncertainty/calibration results, and a simple-model or feature-ablation comparison.
- Train and evaluate the capacity/SOH and backup-time models on real public experimental data; compare formula and ML runtime estimates on common records.
- Add SHAP global/local explanation and integrate it with permutation/feature importance.
- Persist models and preprocessing artefacts and verify reload/inference consistency.
- Complete the Streamlit dashboard including validation feedback, user input, prediction, comparison plots, caveats, and downloads.
- Refine and validate the implemented M9 decision-support thresholds after the primary models are evaluated on real data; retain the non-safety-critical boundary.
- Extend the completed controlled-benchmark unit/integration suite to real-data adapters, model persistence/reload, Streamlit end-to-end behaviour, performance, reliability, manual usability, and residual-risk testing.

10.3 Readiness Assessment

Area	Status	Readiness Statement
Problem definition and scope	Ready	Phase-1 corrections incorporated; core targets and exclusions fixed.
Detailed architecture	Ready	Layered architecture and ten module contracts defined.
Common schema	Benchmark-ready / public adapters pending	CSGB mapping and validation are implemented; NASA/CALCE/Oxford adapters remain source-specific Phase-3 work.
Core ML feasibility	Ready	M5 and M6 execute on the grouped 540/180 CSGB split with aligned holdout evidence.
Formula comparison path	Ready	M4 exact calculation passes; fair operational comparator uses M5-predicted capacity on common records.
Explainability	Partially ready	M5/M6 built-in and ten-repeat permutation evidence is exported and graphed; local/global SHAP case integration remains Phase 3.
Public experimental validation	Not yet complete	Deliberately carried to Phase 3; no real-data metric claim in Phase 2.
Dashboard	Design ready / integration pending	Streamlit selected and interface responsibilities defined.
Testing framework	Executed for controlled benchmark	All modules run individually; 11 automated tests and the package verifier pass. The 34 generated evidence files include 18 PNGs representing 11 distinct views. Real-data, persistence and dashboard tests remain.
Overall Phase-3 readiness	Ready with stated dependencies	Corrected baseline evidence reduces integration risk; real-data adapters, validation, SHAP and dashboard integration are next.

Overall, Revision 2 demonstrates implementation feasibility with auditable baseline data, grouped split integrity, aligned record-level outputs, global importance evidence, M9 class diagnostics, and a fully passing validation suite. The supplied 34 generated evidence files include 18 PNGs representing 11 distinct views. Phase-3 credibility now depends on real public experimental validation, transparent target construction, SHAP local/global case evidence, persistence, and dashboard integration rather than additional duplicate synthetic plots.

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11. Supervisor Review and Approval

Advisor Feedback:

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Supervisor Comments:

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Recommendations:

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Signature: _____ Date: _____